One 61-bit recovery and a memory boundary

This observation report covers every execution of frozen protocol-v9 in the explicit-base end-to-end study. It records bounded public-input recovery and limits, without a scientific verdict, certified speedup, absolute hardware maximum or matched-rho claim. The GPU oracle is generic meet-in-the-middle; the paper's conditional tensor-contraction backend is not implemented on GPU.

Recorded outcomes

Source `e1a38ce4bc586050c3faa67a3ab79f312eb549b5` ran on one RTX PRO 6000 Blackwell Server Edition, pod `nd5qev6envru7z`, created 2026-10-30T15:02:46 UTC. Eight executions produced seven verified recoveries and one memory refusal. Two 62-bit cells were unattempted under the frozen stopping rule: complete both targets at a field size, then stop higher sizes if either failed. V9′s largest complete prescribed rung is 52 bits; its largest individual recovery is 61 bits. The separate v8 panel completed both 60-bit targets. These are different panels.

Bits	Seed	Original index b (s)	Wide index u (s)	Identical x (s)
52	101	29.179238	17.727554	17.724473
52	202	31.296758	22.094780	22.084471
61	101	Not prescribed	0.263632; limited	Not prescribed
61	202	Not prescribed	1537.899248	Not prescribed

Measured cold solve seconds, rounded from `results-v9/rows.csv`. All clocks are shared-host L0 diagnostics, with wall_valid=false. No confidence interval or certified timing ratio is available. Matched 52-bit certificates are identical.

The successful 61-bit case used 1591 relations from 2432 attempts. GPU-oracle time was 1536.244543 s and linear elimination 1.360546 s. The total including fresh CPU certificate replay and scalar-corruption rejection was 1537.918073 s. The other case refused a conservative admission estimate of 85,969,611,408 bytes for 1,788,236,560left rows; it did not perform a failed exhaustive search. The successful case admitted 63,161,668,896 bytes for 1,313,071,104 left rows. These estimates include sort reserve and other allocations, and are not peak VRAM.

Blue: candidate/repeat; green: baseline/CPU; red: timeout/limit. All points retained.

Each point corresponds to a recorded row; the limited row is retained. Run IDs use pod plus field bits, seed and arm, as in `results-v9/export.json`. There is one invocation per prescribed arm/target; x repeats the wide binary.

Contract, validation and custody

The driver accepts certified prime-order curves with `p < 2^62` and `r < 2^63`; hash primes stay in [2p,4p] and below 2^63. The fast Montgomery path remains for smaller moduli; wider hash multiplication uses canonical add/double. Checked scalar/field products use u128, inverse coefficients i128. The field extension changes arithmetic guards, not the relation-finding algorithm.

Command: `bash research/explicit_relations_e2e_20261030/run-boundary.sh`, with EPHEMERAL_OUT=/results, CUDA 13.2/sm_120 and Rust 1.88.0. Seeds 101/202, m=8, tau=2ceil(r^(1/8))+2, `rows <= 2^32`, `work <= 2^34`, chunk 2^20, columns 16384, trials 16. Cold solve caps 180 s at 52 bits, 2700 s at 61 bits and 5400 s at 62 bits; fresh replay 60 s, PARI fixture generation 180 s, worker 18000 s/controller 18300 s. The unchanged admission rule preserves 25% of reported free GPU memory.

Public targets are coordinate lifts; no target scalar is generated or supplied. Each successful certificate passes fresh CPU replay and rejects scalar+1. Curve equation, primality and the unique Hasse-interval order witness are checked. All 19 Rust tests, host/full/cached/compact device checks, Clippy, maximum-cap headers and full/compact/wide query memcheck, racecheck and synccheck passed. Compile/preflight and fixture generation are outside cold solve timing. Cold solve includes startup, checks, all relation trials, elimination, extraction, in-process replay, serialization and teardown; the additional fresh replay and negative control are included in total audit. CPU user/system time and measured peak VRAM were not captured.

Frozen protocol hash: `bf1e4a2dddbfdadb2cb8de32a0c659bbcdd528cc2851c63864c66d009f4e5142` Raw archive SHA256: `cf5f4094f472ab688c11d14823df598ce6141d8ae2ee1ad70513e7881a96d0f4` Archive: `s3://setec-astronomy-artifacts/validation/ecbench-explicit-x/e2e-20261030-v9/results.tgz`. Raw records were committed/pushed in `6e850f7dfc14` before this report. The archive manifest hashes every member; certificate-audit.json records native replay and rejection outputs. Those manifests do not replace replay. Worker/controller exited zero, archive digest matched, deletion was verified, and a later provider listing independently confirmed the pod's absence.

Schema, flow and file hashes pass. The frozen native analyzer and strict frontier audit pass; the latter rejects eight coverage/status/input mutations. The canonical protocol checker retains five rectangular-coverage findings: 18 assumed cells versus 8 actual, u covers 2 rather than 3 sizes, b and x each cover 1 rather than 3, and arm instance sets differ. This is not a full canonical checker pass or independent scientific validation. Details are retained in `results-v9/CHECKER-NOTE.md`; no failed row was excluded.

Literature and remaining implementation work

The checked Fp64 arithmetic contract, the supplied paper's inverse-hash construction, HAC algorithms 2.107/2.142 and PARI ellcard documentation informed the width extension. See `gpu/explicit-relations/WIDE-HASH.md` and PARI's elliptic-curve documentation. V10 was frozen but never executed. A separate v11 protocol tests exact 12-byte rows and queried CUB sort scratch with distinct 52-bit controls before larger probes. Its planned results are not included here. Neither this report nor a merged PR settles the paper's conditional backend or the hardware maximum.